Thermic radiation of Leslie's cube and a black body

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We try to figure out, whether *Leslie's cube* follows T^4 according to *Stefan-Boltzmann's law*. With a reference temperature of $T_0 = (291.39 \pm 0.50)$K, the exponents $b = 4.06261 \pm 0.00021$ (black side), $b = 4.06087 \pm 0.00019$ (white side), $b = 3.98776 \pm 0.00019$ (matt side), and $b = 3.76106 \pm 0.00062$ (polished side) were found. For a different experimental setup, using a black body, and higher temperatures, $b = 4.020 \pm 0.013$ was observed. $\lambda_{room} = (9.885 \pm 0.034)\mu$m up to $\lambda_{Leslie} = (8.323 \pm 0.024)\mu$m is the wavelength range observed. The transmission of different filter materials is: $(100.0 \pm 4.1)\%$ (no filter), $(0.00 \pm 0.38)\%$ (nickel-plated), $(2.06 \pm 0.44)\%$ (window glass), $(46.7 \pm 2.1)\%$ (silicon), $(78.6 \pm 3.3)\%$ (NaCl), and $(1.36 \pm 0.42)\%$ (infrared filter). The heat of real surfaces was found to not exactly follow *Lambert's law*, because they seem to prefer emission in the perpendicular direction.

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1 Introduction

A deep understanding of radiation was only achieved by *Max Planck* in 1900 through his introduction of quantum mechanical concepts into electrodynamics. Still, we can describe plenty of experimental data with classical aspects. The heat radiated by an object is heavily dependent on its surface. A box with black walls on the inside and only a small hole for the radiation to pass through is usually used to approximate the theoretical concept of a black body, which has perfect absorption. A *Leslie cube* can be used to compare different surfaces. It is filled with water of consistent temperature and has different materials and/or surfaces to all sides. The amount of heat radiated by a black body follows T^4 according to the *Stefan-Boltzmann law*. How good this approximation holds in reality, is tested in the first part of this experiment. Real materials do not just absorb infrared light, but rather let part of it pass through. The degree to which they absorb the considered wavelengths is measured for different materials in the following part of this paper. Lastly, we took a look at the direction in which the black body sends out its radiation. According to *Lambert*, this directional dependency should follow the cosine function.

2 Stefan-Boltzmann law

The *Stefan-Boltzmann law* describes the radiated power P of an area A as a function of temperature T.

$$P = A\sigma T^4 \tag{1}$$

σ is called the *Stefan-Boltzmann constant*. This law applies to the black body and can be derived from the radiation law according to *Planck*. In the following subsections, we will check experimentally if the *Stefan-Boltzmann law* applies to the sides of a *Leslie cube* and an experimental approximation of the black body.

2.1 Examination of Leslie's cube

The experimental setup is shown in fig. 1. In order to investigate the dependence of thermal radiation on the surface properties and chemical composition of various materials, a so-called Leslie cube was used. It is rotatably mounted so that the various sides can easily be adjusted during the experiment. A black, a white, and two nickel-plated surfaces, one matte, the other polished, were available for selection. The thermostat *Lauda E100* heats or cools the cube to the desired temperature, which was increased from 16°C to 90°C. Via a thermopile produced by *Kipp & Zone* the emitted thermal radiation of the respective side is detected in dependence on the adjusted temperature. Due to the

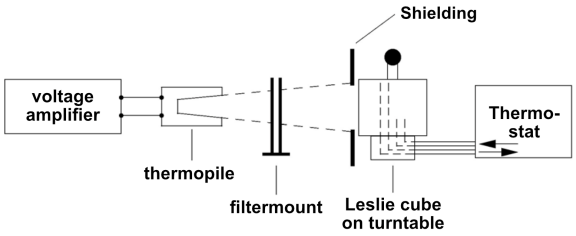

Figure 1: The thermostat regulates the temperature of Leslie's cube, which is rotatably mounted so that the four different sides can be examined. Its emitted heat radiation is measured via the thermic element, whose signal is amplified. A filter can be mounted in the radiation's path. This is only done in section 3. [1]

Seebeck effect, the radiation power is directly proportional to the voltage, which was digitally output by the multimeter *Agilent 34405 A*. Due to this measurement method, eq. (1) must be corrected by the reference temperature T_0 with which the thermopile operates. This is not exactly the ambient temperature of 20°C, but the temperature at which the voltage is 0. To obtain this value, it was determined for all the different sides and averaged. For this purpose, the first four measured values were linearized and the intersection with the axis on which the temperature is plotted was read off.

$$T_0 = (291.39 \pm 0.50)\mathrm{K}$$

With this result, the measured data can be fitted with the following equation.

$$U \sim P = k \cdot (T^b - T_0^b) \tag{2}$$

To determine the proportionality factor k, the corresponding value was read from the fit of the black side. The reason for this is that this side should be the best approximation for a black body. Since k is a characteristic quantity of the thermopiles surface, with which the measurements were taken, this cannot influence the remaining exponents as an additional parameter.

$$k = (1.92 \pm 0.54) \cdot 10^{-10} \frac{\mathrm{mV}}{\mathrm{K}^4}$$

Graph 2 shows the voltage, proportional to the radiation intensity, as a function of temperature with the corresponding fit for all four measurement series. It can clearly be seen that the white and black surfaces exhibit the same functional behavior, i.e., they emit almost the same thermal radiation. This means that although the two colors have different emission and absorption in the visual spectrum, they behave like the same color in the selected temperature and infrared range. The matt nickel-plated side also has a high emission, as the roughness ensures a big surface area. The polished

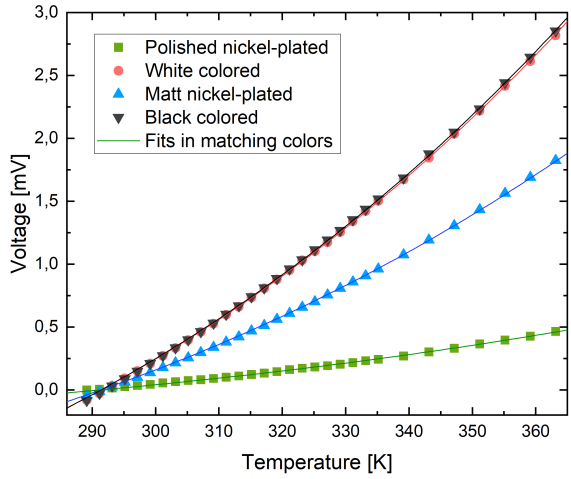

Figure 2: The measured voltage as a function of temperature is shown for all four sides of the Leslie cube. The surfaces are white (red), black (gray), matt nickel-plated (blue), and polished nickel-plated (green). In addition, a fit was made using $U \sim k \cdot (T^b - T_0^b)$.

nickel side, on the other hand, has a much higher reflectivity and emits much less because the surface finish is very smooth. From the fits, the following values were obtained for the exponents, which were sought to check the *Stefan-Boltzmann law*:

surface	exponent b
black	4.06261 ± 0.00021
white	4.06087 ± 0.00019
matt nickel-plated	3.98776 ± 0.00019
polished nickel-plated	3.76106 ± 0.00062

Table 1: Determined exponent b for all four sides of the *Leslie cube*

Although all results are close to the theoretically expected value $b = 4$, none of them overlap within their error bars. This is because the accuracy of T_0 and k cannot be judged from the fit. It can also clearly be seen in fig. 2, that the white and black sides behave in the same way. Accordingly, both surfaces can fulfill the approximation of a black body and satisfy the *Stefan-Boltzmann law*. In contrast, the shiny surface deviates significantly more and the approximation as a black body cannot be confirmed experimentally. The reason for this is that part of the heat radiation of the cube is absorbed by the material through lattice vibrations. In general, it can also be said that systematic deviations arise due to the heat radiation from the environment. A shielded setup would have been more

optimal. The heating also causes the reference temperature T_0 to increase with temperature. This leads to an underestimation of the radiation power. This is especially relevant in the next section as the maximum temperature is significantly higher. To verify the fits, the residuals of all measurements are listed in the appendix (fig. 7). They show that the previously mentioned sources of error could not be taken into account and thus have an effect on the quality of the fits.

2.2 Examination of a black body

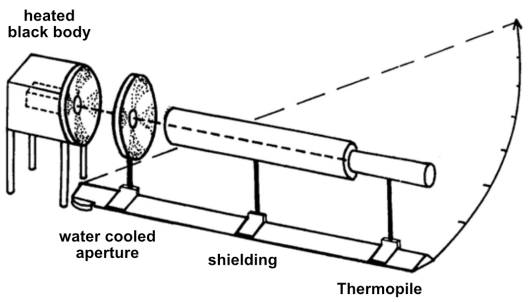

Figure 3: The heated black body emits thermic radiation through water-cooled orifice plates, which is measured via a thermopile. Furthermore, the experimental setup can be rotated so that the angular dependence can be investigated in section 4. [1]

The measurement and evaluation are analogous to the previous part of the experiment. Instead of the *Leslie cube*, the thermal radiation of a model black body (fig. 3) is now investigated. For this experiment, the orientation is 0°. During the continuous heating of the black body, the voltage on the multimeter *Agilent 34405 A* was read off in 5°C steps. The measured temperature dependence is shown in fig. 4. Since the thermopile is the same model as the one in the previous subsection, the same T_0 value was used to fit the data according to eq. (2). From this, the following exponent can be determined:

$$b_{black\,body} = 4.020 \pm 0.013$$

This result as well cannot be reconciled with the theoretical value within the error. However, it should also be mentioned that the error is underestimated since the quality of the parameters is not included, and also not the systematic errors of the environment and the temperature dependence of T_0. This can also be seen in the residuals in fig. 8. Taking this into account, the Stefan-Boltzmann law fits the oven used, which can therefore be considered a good model of a black body radiator.

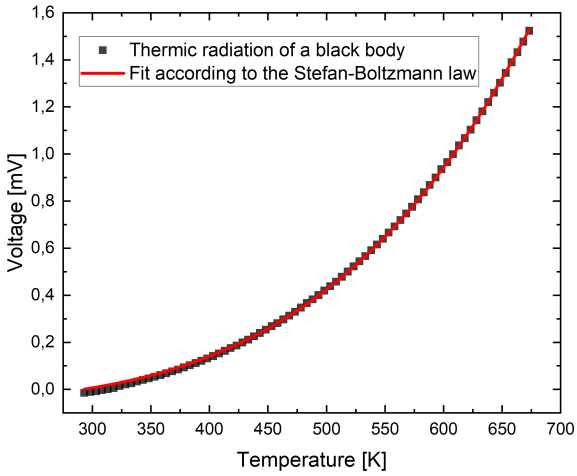

Figure 4: The heat from an oven in grey was measured for different temperatures, a red fit through the points is used to compare to the *Stefan-Boltzmann-law*

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3 Infrared transmission of different materials

Again, we use fig. 1 but now with different filters. We compare no filter, nickel-plated, window glass, silicon, and NaCl. The expectation is that no filter gives the highest amplitude while all other filters have material-dependent transmission. For the measurements, a temperature of 75°C was selected. The room temperature was around 20°C. The measured transmission of the nickel-plated filter was used to correct the voltage offset because no radiation would have been able to pass through. After inserting the filter-type x, fraction Q of the Intensity without filter remained:

$$Q = \frac{U_x}{U_{nofilter}} \tag{3}$$

The error is calculated from the Agilent's $2.5\,\% + 0,8\,\mathrm{mV}$ DC error [2] in the 100mV sensitivity range and *gaussian propagation of uncertainty*. The silicon filter was oriented with the mirror-like surface towards the opening of *Leslie's cube*.
The results from table 2 show that the absorption or transmission of electromagnetic radiation, especially infrared radiation, depends on the structure and properties of the material. Infrared radiation is energetically in the same range as the vibration of molecular bonds, i.e. absorption leads to an excitation of the bonds. The material dependency is a result of the different bond types, since these characterize the necessary energy. Due to the attractive and repulsive interactions between atoms in solids, the optimal bond distance in the molecule is at the minimum of the potential. A molecule in an electric field aligns itself with its dipole moment along the electric field and increases

x	Q
No filter	$(100.0 \pm 4.1)\%$
Nickel plated	$(0.00 \pm 0.38)\%$
Window glass	$(2.06 \pm 0.44)\%$
Silicon	$(46.7 \pm 2.1)\%$
NaCl	$(78.6 \pm 3.3)\%$
Infrared filter	$(1.36 \pm 0.42)\%$

Table 2: Transmission in the infrared spectrum for different materials. The mirror surface of silicon was turned towards the oven.

its bond distance. Electromagnetic waves lead to oscillations or rotations of the bonds in groups. In an amorphous structure, such as window glass, a lot of energy is absorbed by the excitation of the energy bands. For this reason, the transmission of thermal radiation with $Q_{window} = 2.06\,\%$ is very low. Molecules without a permanent dipole moment would accordingly show no absorption. Nevertheless, the transmission for silicon, a material with covalent bonds, that is without a permanent dipole moment, is only $Q_{silicon} = 46.7\,\%$. In addition to the mechanical explanation, the band pressure must now be examined in more detail. The band gap for silicon is $E_{gap} = 1.12\,\mathrm{eV}$ [5]. However, this would again lead to the result that the material would have to be transparent for radiation, since the transition from conduction to valence band is unlikely at these radiation energies. The supposed absorption could also be due to a reflection of the thermal radiation, which would also inhibit the transmission. NaCl has an even wider energy gap with $E_{gap} = 8.78\,\mathrm{eV}$ [7], meaning the material should also be permeable. Stronger chemical bonds and atoms of smaller mass cause absorption maxima at high energies in the IR spectrum. Only a part ($Q_{NaCl} = 78.6\,\%$) of the infrared radiation is transmitted. Infrared filters are used to sort out the long-wave spectral range. Therefore, its transmission coefficient $Q_{infredfilter} = 1.36\,\%$ is the lowest. [3], [4], [5], [6]
The wavelength peak λ_{Peak} of Planck's thermic radiation with which the filters were hit can be calculated with Wien's displacement law eq.(4).

$$\lambda_{Peak} = \frac{b}{T} \tag{4}$$

$b \approx 2,897771955 \cdot 10^{-3}\,\mathrm{m} \cdot \mathrm{K}$ is *Wien's displacement constant* and T is the absolute temperature in Kelvin. From this, we can calculate the wavelengths λ at which Planck's radiation law has its maxima at room Temperature and at the temperature of *Leslie's cube*.

$$\lambda_{room} = (9.885 \pm 0.034)\,\mu m \tag{5}$$
$$\lambda_{Leslie} = (8.323 \pm 0.024)\,\mu m \tag{6}$$

Both these wavelengths lie firmly inside the infrared region of the em-spectrum. All the radiation hitting the thermic element in this experiment should have a wavelength in between these two values.

4 Lambert law

The angular distribution of the thermic radiation field of a black body is described by the *Lambert law* (eq. (7)).

$$J = J_0 \cdot \cos(\varphi) \tag{7}$$

This comes from the fact that the area visible to the observer depends on the angle of observation. If a source of radiation does not follow eq. (7), it is called a *non Lambert radiator*, which usually means the heat output is concentrated along the direction perpendicular to the area. To approximate a black body, the oven from Chapter 2.2 is used again (compare fig. 3). The oven's temperature was set to $350 \pm 10\,^\circ\mathrm{C}$, and the intensity of the radiated heat was measured as a function of angle φ between the detector and the normal direction of the opening through which the black body can be seen. The temperature error was estimated generously because the thermometer is already pretty unprecise, and significant shifts in the oven's temperature during the experiment could not be avoided. The angle was measured in 5° steps up to 50°. In fig. 5 the measured

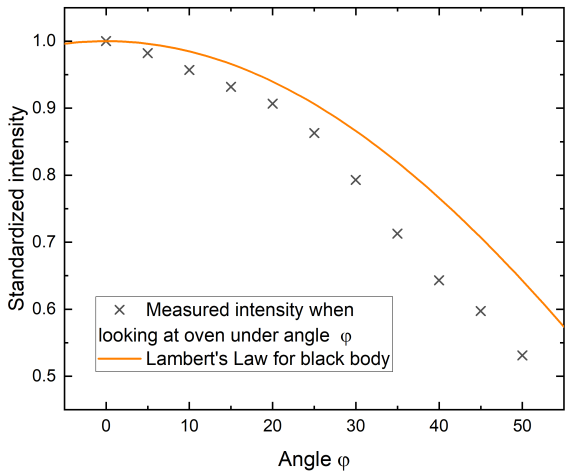

Figure 5: The measured dependency of the black body's heat intensity (grey crosses) on the observation angle φ is compared to *Lambert's Law* in orange.

intensity is plotted and compared to the expected results for a perfect black body (eq. (7)). The measured

values are scaled in order to get value 1 for an angle of $\varphi = 0°$. While the ruff shape of the measured values looks like a cosine, the slope is much greater. This is the normal behaviour of non-ideal black bodies. They radiate primarily in the direction of the opening. In fig. 6 the intensity is compared to the cosine of the angle. Theoretically a straight line is expected. If a line is fit through all points except the first two the deviation of the points from this line is quite small. From this, we can follow that Lambert's Law is fulfilled for larger angles, while the radiation directly in front of the oven does not follow this law.

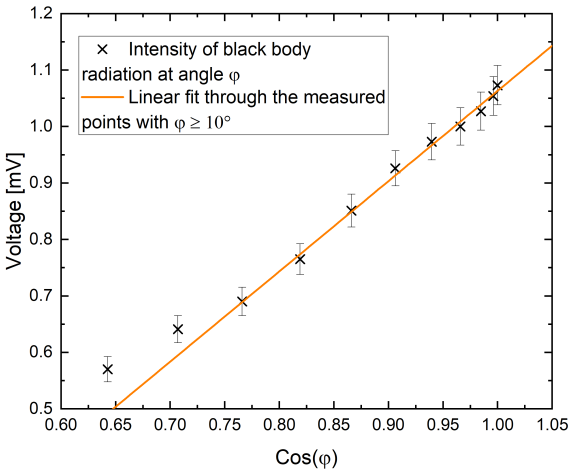

Figure 6: When an oven was observed under the angle φ the intensities plotted in grey were observed. A linear orange fit goes through the points with $\varphi \geq 10°$

5 Summary

In the first part of our paper, we take a look at how well the radiation from the different surfaces of *Leslie's cube* follows the T^4 temperature dependency predicted by *Stefan and Boltzmann*. The temperature relative to which the thermic element takes its measurements was determined to be $T_0 = (291.39 \pm 0.50)$K. The exponent for the black side of *Leslie's cube* is $b = 4.06261 \pm 0.00021$, the white side has $b = 4.06087 \pm 0.00019$, the matt side follows an exponent of $b = 3.98776 \pm 0.00019$, and the polished nickel-plated side has $b = 3.76106 \pm 0.00062$. We, therefore, conclude, that *Stefan-Boltzmann's law* can be used to predict the intensity of radiation of the tested materials. For materials like polished nickel, adjustments should be made if an exact power law is required. In the next part, an oven with a bigger temperature range was used to examine the T^4 power law. An exponent of $b = 4.020 \pm 0.013$ was found which also strengthens *Stefan-Boltzmann's law*. The wavelengths used in this experiment are between $\lambda_{room} = (9.885 \pm 0.034)\mu$m and $\lambda_{Leslie} = (8.323 \pm 0.024)\mu$m, hence all within the infrared part of the spectrum. If different materials are brought into the heat ray and the measured intensity is compared to the intensity with no obstructions, we can figure out how effective different materials are as infrared filters. The transmission of no filter was observed as $(100.0 \pm 4.1)\%$, the nickel-plated metal has transmission $(0.00 \pm 0.38)\%$, window glass transmits $(2.06 \pm 0.44)\%$ of radiation, silicon $(46.7 \pm 2.1)\%$, NaCl $(78.6 \pm 3.3)\%$ and the infrared filter with $(1.36 \pm 0.42)\%$ absorbs almost all of the radiation. If the black body is not observed head-on, but rather at an angle, the intensity should follow *Lambert's law*. The radiation of the oven from above was observed to follow this law for bigger angles of observation, but a bigger intensity part than expected is measured directly in front of the oven.

References

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6 Appendix

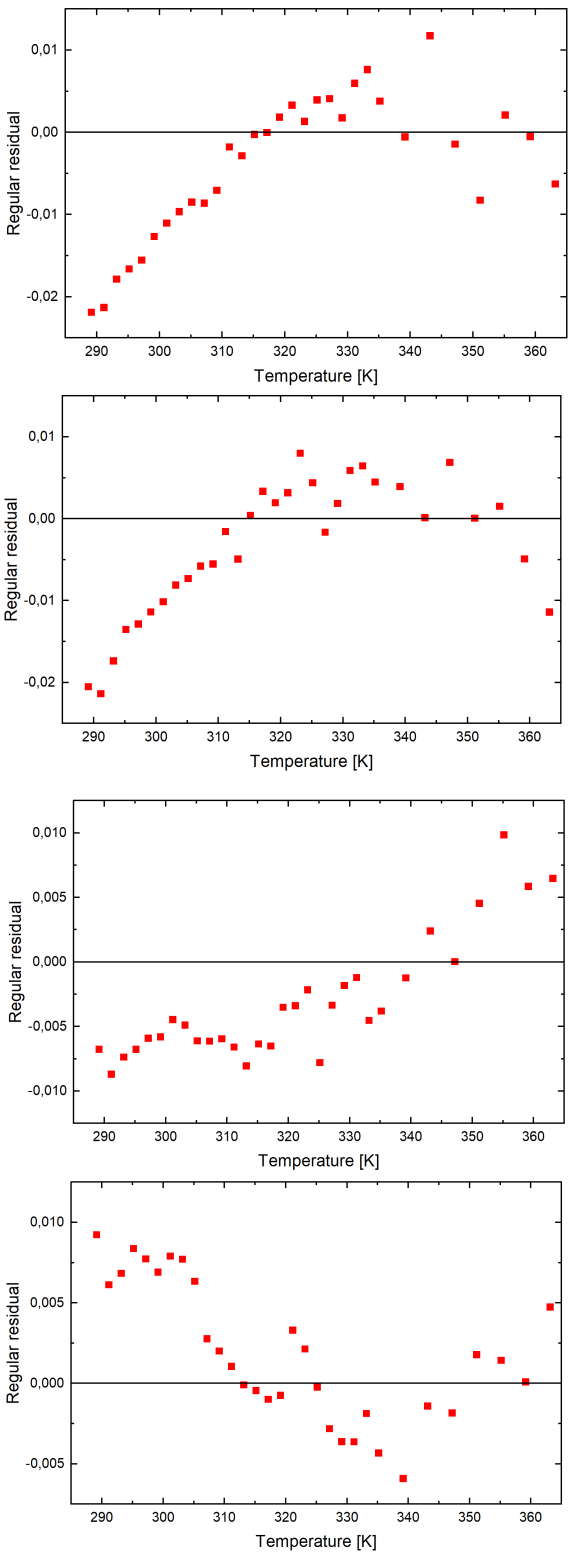

Figure 7: Residuals of the fits from the four measurement series to Leslie's cube, in the following order: black, white, matte and polished nickel-plated

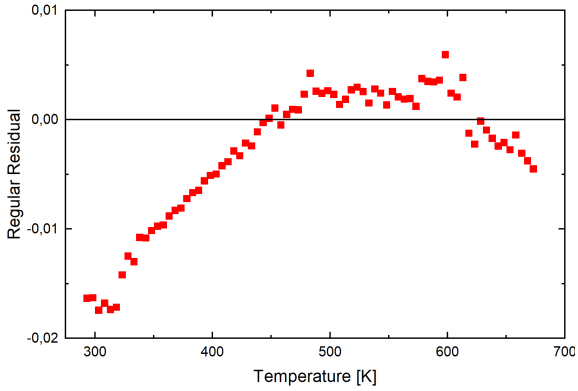

Figure 8: Residual of the polynomial fit for the heated black body.